American options: policy evaluation and fitted Q-learning

Bermudan put, 50 exercise dates: Longstaff–Schwartz vs fitted Q-learning

Team 9 · Jaskaran Kalra, William McDonnell

Q1: the value function

(1) $V^a_n(s,x) = E[\Sigma_{m \geq n} \beta^{m-n} x_m a_m g(S_m) \mid S_n = s, x_n = x]$

(1) Recursion: $V^a_n = x\, a_n\, g(s) + \beta\, E[V^a_{n+1}(S_{n+1}, (x - a_n)^+)]$

Why $x = 0$ stays 0, so an exercised option has value 0: it pays once

(2) $C^*_n(s) = \beta\, E[V^*_{n+1}(S_{n+1},1) \mid S_n = s]$, $V^*_n = \max(g, C^*_n)$

Why S is Markov: past prices add nothing about S_{n+1}, so using history cannot beat V^*

(3) Two-period check

$q = 0.625$

Up (120): $C_1 = 1.43 > g = 0 \rightarrow$ continue

Down (80): $C_1 = 15.24 < g = 20 \rightarrow$ exercise

$V_0 = 7.9932$

Why alive differs: down–up was already exercised at 80; up–down is still alive and pays 4 at the end

Q2: Longstaff–Schwartz

(1) Backward over dates; fit on in-the-money paths only; then set $\hat{\tau} = n$ where rule (7) says exercise

(2) Date 0: no regression, use the average cash flow; fitted value = max(average, $g(S_0)$)

(3) Why cash flow one realized draw; its average given S_n is the continuation value, so no fitted value is fed back

(3) Why future allowed in training labels, but not in evaluation: that would be look-ahead

(3) Why date n the response is compared with $g(S_n)$, paid at date n

Response (8)

$$Y_n = \beta^{\hat{\tau}-n}\, g(S_{\hat{\tau}})$$

$\hat{\tau}$ = first exercise date of the already fitted future policy (starts at N)

Both methods: basis 1, z, …, z^d with $z = s/K - 1$, $d = 2, 4$; least squares by lstsq; clip to [0, K]; exercise iff $g > 0$ and $g \geq \hat{C}_n$

Q3: fitted Q-learning

(1) Backward over dates; fit on all paths; freeze $\hat{V}_{n+1}$ before fitting date n

(2) Date 0: clipped average of targets vs $g(S_0)$; actions from rule (7); evaluated on fresh paths

(3) From (9) $Q^*(\text{continue}) = \beta\, E[V^*_{n+1}]$: replace V^* by $\hat{V}$ and the expectation by regression

(3) vs (8) one step realized, the rest estimated ($\hat{V}$); in (8) the whole cash flow is realized

(3) Why all paths an out-of-the-money stock can still fall below K, so $\hat{V}_n = \hat{C}_n > 0$ there and feeds earlier targets

Targets (10)–(12)

$$\hat{V}_N = g$$

$$Y_n = \beta\, \hat{V}_{n+1}(S_{n+1})$$

$$\hat{V}_n = \max(g, \hat{C}_n)$$

Errors in $\hat{V}$ and the max carry back through all dates

Q4 (1): checks and the twelve policies

policy	fitted V_0	seed 17	seed 29	seed 43
LSM, d = 2	5.98–6.10	6.059	6.052	6.056
FQ, d = 2	7.85–7.95	5.852	5.851	5.849
LSM, d = 4	5.99–6.10	6.078	6.070	6.077
FQ, d = 4	6.90–7.01	6.019	6.005	5.978

Independent policy values (100,000 evaluation paths, SE ≈ 0.02–0.03). Benchmark 6.0786.

Why FQ fitted values are far above its policy values (≈ 35–80 SE): regression errors and the max carry back through all dates; not Monte Carlo noise

Checks

Two-period: evaluator gives 7.9932 = V_0

European: 5.588 vs Black–Scholes 5.574 (z = 0.51)

Reference: our CRR tree reproduces 6.0784 (2500 steps)

Below bound: all 12 policy values are below the benchmark

Q4 (2)–(3): paired differences and decisions

(2) FQ − LSM, d = 2: -0.21, -0.20, -0.21

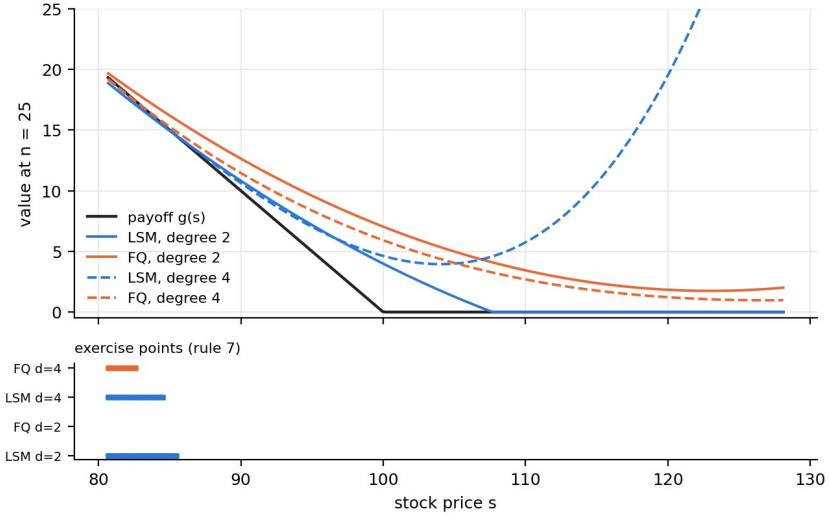
(2) FQ − LSM, d = 4: -0.06, -0.06, -0.10

(2) All six 95% CIs are below 0: LSM is better

Why paired same evaluation paths cancel common noise (SE 0.009–0.015 vs ≈ 0.034 unpaired)

(2)–(3) degree 4 narrows the gap and raises both methods' policy values, much more for FQ

Why small near the crossing g ≈ C, a wrong decision loses only |g − C| ≈ 0



Seed 17, n = 25. Exercise for s ≤ 85.4 (LSM d = 2), 84.5 (LSM d = 4), 82.6 (FQ d = 4); FQ d = 2 never exercises

Q4 (4): what the evidence supports

Bound Given the training data, a frozen policy is admissible, so $E[H \mid \text{data}] \le V^*_0 \approx 6.079$ (Part 1)

Test means can still exceed V^*_0 by noise (about one SE ≈ 0.02)

Fitted values are not payoffs of a policy: LSM is in-sample, FQ has max and basis bias, so neither must be below V^*_0

SE measures only evaluation Monte Carlo noise for a fixed policy

SE omits training-seed variation, basis/degree error, and reference error

Result: LSM d = 4 is within 0.009 of the benchmark for every seed; fitted Q is lower in all six paired comparisons, and its fitted values overstate what its policies earn.